

CO 3 ASSESSMENT TOOL 2

30. FPGA-Oriented Polyphase Filter Simulation

Simulate an FPGA-oriented polyphase filter architecture and evaluate its hardware efficiency.

(25) (BL5) (WK4)

Given Data:

- FIR Length = 48 taps
- Decimation Factor = 4

Tasks:

1. Design the polyphase filter.
2. Simulate the FPGA-oriented architecture.
3. Estimate multiplier usage.
4. Estimate memory requirements.
5. Compare with direct architecture.
6. Recommend the most suitable implementation approach.

SOLUTION:

INTRODUCTION:

A polyphase filter is an efficient implementation of a Finite Impulse Response (FIR) filter used in multirate digital signal processing. It divides the filter into multiple sub-filters (phases), reducing the number of computations during decimation or interpolation. In FPGA implementation, polyphase filters improve hardware efficiency by reducing multiplier usage,

memory access, and power consumption while maintaining the required filtering performance.

PROGRAM:

% FPGA-Oriented Polyphase FIR Filter Simulation

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clc;

clear;

close all;

N = 48; % FIR Filter Length

M = 4; % Decimation Factor

% Design Low-pass FIR Filter

h = fir1(N-1,1/M);

% Polyphase Decomposition

poly = reshape(h,M,[]);

disp('Polyphase Coefficients:');

disp(poly);

```
% Input Signal
fs = 1000;
t = 0:1/fs:1;
x = sin(2*pi*50*t) + 0.5*sin(2*pi*200*t);

% Direct FIR Output
y_direct = filter(h,1,x);

% Decimated Output
y_poly = downsample(y_direct,M);

% Plot
figure;
subplot(2,1,1);
plot(x);
title('Input Signal');

subplot(2,1,2);
plot(y_poly);
title('Polyphase Decimated Output');

% Resource Estimation
fprintf('Filter Length = %d\n',N);
fprintf('Decimation Factor = %d\n',M);
```

```
fprintf('Multipliers Required = %d\n',N/M);
```

```
fprintf('Memory Requirement = %d Coefficients\n',N);
```

TOOLS USED :

- MATLAB/Simulink (for filter design and simulation)
- Xilinx Vivado / Intel Quartus Prime (FPGA implementation)
- HDL (Verilog/VHDL) for hardware description
- DSP System Toolbox
- FPGA Development Board (optional for hardware testing)

RESULT:



Figure 1 X

Figure 2 X

